Automated Classification of Fresh and Rotten Fruits Using ResNet50 for Enhanced Food Quality Control and Waste Reduction1

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Abstract— Effective classification of fresh and rotting fruits is essential in the agricultural and food industries for food safety, quality control, and waste minimising purposes. The ResNet50 convolutional neural network (CNN) model is investigated in this work to automatically classify fruit freshness, therefore tackling the difficulty of differentiating between fresh and rotting fruits. Comprising 1,655 images from six categories fresh peaches, fresh pomegranates, fresh strawberries, rotten peaches, rotten pomegranates, and rotten strawberries the ResNet50 model achieves an overall accuracy of 95%. Strong performance across various metrics including precision, recall, and F1-score is shown by the model, which distinguishes fresh pomegranates and rotten strawberries especially with very great accuracy. Training and validation trials show strong generalization with low overfitting; the confusion matrix points up areas needing work and shows how well the model can fairly classify fruits. Emphasizing the ability of automated fruit classifying systems in improving supply chain efficiency and lowering food waste, this work offers a scalable way for fruit quality monitoring.

Keywords— Fruit Classification, ResNet50, Convolutional Neural Network (CNN), Fresh vs. Rotten Fruit, Image Classification, Food Waste Reduction, Deep Learning.

I. INTRODUCTION

Fresh and rotting fruit classification is a useful application in the food and agricultural industries attracting increasing interest because of its effects on quality management, food safety, and sustainability [10]. Fruit is fairly perishable, hence it naturally breaks down under the influence of temperature, humidity, and microbial activity. This makes quick and precise diagnosis of rotting essential to reduce food wastage and guarantee the quality of goods getting to customers [11]. Automating the fresh and rotten fruit classification process takes front stage given the worldwide focus on lowering food loss and streamlining supply networks. Computer vision methods allow one to effectively discriminate between fresh and rotten fruit, therefore improving processes in inventory control, sorting, and packaging. Fresh and rotting fruits differ visually mostly in color, texture, mold presence, and soft spot count; these are usually minor but important alterations [12]. A rotting peach could show discolouration, obvious mould, and texture changes like wrinkling or softening; a fresh peach is vivid in colour, firm to the touch and blemish-free. These elements are very important for classification since freshness is mostly determined by the visual deterioration of fruit [13]. Large amounts of fruit must be manually assessed, which takes time and increases human mistake-prone behaviour leading to inefficiencies. Thus, by giving precise and scalable approaches for monitoring fruit quality, automated fruit classification systems—powered by machine learning models offer a solution to these difficulties. In this sense, classification is the method of grouping fruits according to their outward look [14]. Not only is this essential for quality control but also for besting storage conditions and waste minimization. Being able to rapidly identify and separate fresh from rotting fruit guarantees that only premium product reaches the consumer in large-scale operations such supermarkets, farms, and supply chains, therefore improving customer happiness and lowering the risk of foodborne diseases [15].Good classification affects the economy too. Should not be discovered early, rotten fruits could influence other surrounding fruits, hastening their disintegration and thereby increasing losses. Correct fruit classification helps companies to reduce financial losses and help to balance waste of food [16]. Correct fruit classification can help to improve post-harvest management, therefore guaranteeing ideal use of food and more effective distribution of it across the supply chain in a world when food security is an increasing issue.

II. LITERATURE

Maintaining fruit quality and lowering food waste have made detecting rotten fruits rather important in the agricultural sector. Unlike people, machines have no restrictions, hence automation in this field is rather helpful. Reducing human labour, cutting production costs, and accelerating fruit defect identification is the main emphasis of a suggested method. Should flaws in fruits go unnoticed over time, rotting fruit might contaminate fresh ones, therefore causing additional spoiling. [1]. Starting with data collecting, multiple linked stages are engaged to improve the categorization process: After that, methods including colour uniformity, image resizing, augmentation, and tagging help to preprocess the data. Three publicly available datasets have been used for detailed simulations yielding positive findings with accuracies of 98.2%, 99.8%, and 99.3%, respectively. Furthermore computationally efficient, the suggested approach averages 8 milliseconds to provide the final classification outcome [2]. In many different sectors, including retail, where it enables suppliers in supermarkets to identify fruit species and decide pricing, fruit categorization is crucial. Furthermore, classification helps to identify fresh fruit during the exporting process, which is absolutely vital for preserving financial viability. Applications for autonomous agricultural robots and smartphone applications for the identification of several fruit species find place in fruit categorization systems. In one work, rather than using conventional CNN architectures, a total of 5,658 fruit photos from 10 classes were employed to show the possible of transfer learning in CNN models for image classification. Based on test results [3]. The food sector also gives great weight to the need of checking fruits before they are sold to customers. Manual procedures involve time and

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energy, so human inspection suffers from discrepancies and mistakes. Using deep learning methods—more especially, CNNs—a suggested approach extracts features and classifies rotting fruits. For this instance, oranges, apples, and bananas fall under a validation accuracy of 98.89%. With an average classification time of 0.2 seconds per fruit image [4], the model spent over 212.13 minutes overall in training. This work also presents a deep learning method using photos to identify fresh from rotten fruits, therefore tackling the classification difficulty with CNNs. Labeled photos of fruits at varying degrees of freshness comprised a varied dataset. Resizing and normalizing the photos among other preprocessing tasks guaranteed consistent input for the model. Starting from a pre-trained CNN architecture, the model may be adapted to the particular job of fruit categorization, therefore offering a suitable solution for quality control in the food industry [5]. In another work, a unique method for categorizing fruit freshness was developed combining CNNs with Support Vector Machines (SVMs). The study's goal was to automate the often human error-prone manually labor-intensive fruit freshness determination process. Pre-trained ResNet50 model was trained using the dataset comprising photos of fresh and rotten apples, bananas, and oranges. CNN model extracted high-level images; these were then fed into an SVM classifier for final classification. Among other measures of the models' performance were accuracy, precision, and F1 score [6]. Fruits should be accurately categorized depending on freshness considering their essential supply of nutrients, vitamins, and fibers.Various image processing methods have helped many fruit kinds be categorized and identified. Using neural networks to investigate the features [7] helped one to ascertain the quality of the fruits by means of data. In another work, a fruit image classification model able to differentiate fresh and rotting fruits was developed using transfer learning methods. By means of hyperparameter adjustment of the best-performing architecture, the model performance was improved; the outcomes show a rather strong classification model [8]. Many fruits, once picked, go through several phases of deterioration if improperly handled. The problem of fruit rot detection was investigated as a binary classification challenge. CNN models were trained and tested using a collection of fresh and rotten fruit photos, therefore proving its relevance in preventing rotting throughout the transportation and packing process [9].

III. INPUT DATASET

This study makes use of 1,655 images overall split into six categories based on freshness and type of fruit using the open-source Kaggle platform. Among these abundant fresh peaches, fresh pomegranates, fresh strawberries, rotting peaches, rotting pomegranates, and decaying strawberries. The particular split comprises of 250 fresh peach photographs, 311 fresh pomegranate images, 250 fresh strawberry images, 343 rotting peache images, 250 rotting pomegranate images, and 251 rotting strawberry images. This varied collection fit for creating a strong classification model since it offers a comprehensive picture of both fresh and rotting fruit. Three separate subsets totaling 70% for training, 15% for validation, and 15% for testing have been created from the dataset to guarantee efficient training and model evaluation. The ResNet50 model is trained to differentiate fresh from rotting fruit using the training set that which makes up majority of the dataset. Using reference to the validation set, the hyperparameters of the model are changed such as to prevent overfit during the training process. At last, the test set provides an independent assessment criterion to investigate model generalizing capacity and performance. To guarantee uniformity in size and scale, every image in the collection passes preprocessing techniques including scaling and normalizing. This standardization ensures consistency during the training phase by helping the model focus on spotting special traits between the fresh and rotting fruit images. By means of this well-organized and labeled dataset, the research aims to get accurate classification results so benefiting applications in agricultural and food quality management. As Shown in Fig.1.

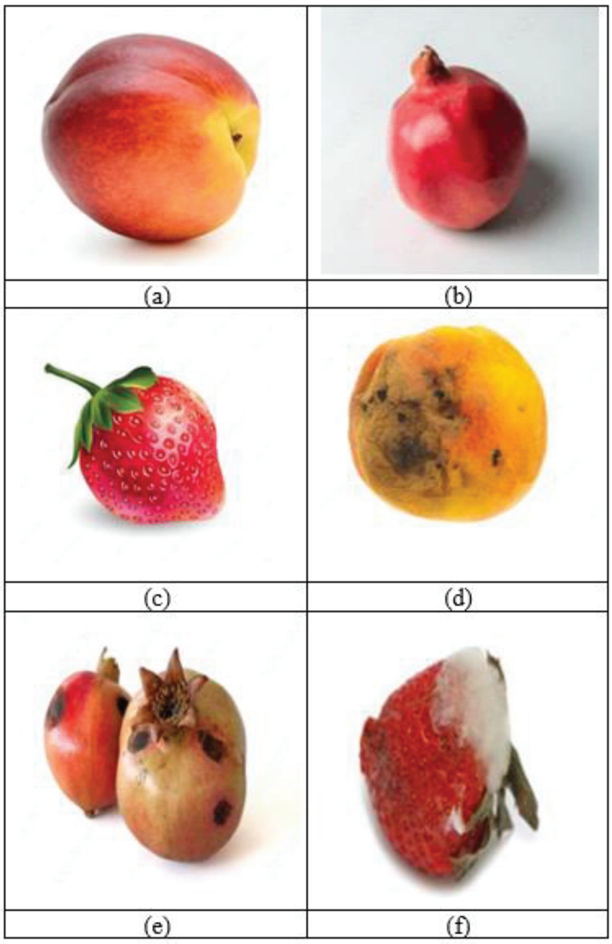

Fig. 1. Dataset image for (a) fresh_peaches_done, (b) fresh_pomegranates_done, (c) fresh_strawberries_done, (d) rotten_peaches_done, (e) rotten_pomegranates_done, (f) rotten_strawberries_done.

IV. PROPOSED METHODOLOGY

The suggested study strategy consists of four primary phases: data preparation, model creation, training and validation, and evaluation. Raw images of fresh and rotting fruits are scaled, standardised, and improved in the first phase, data preparation, to ensure consistency and motivating model generalisation. Using a custom classification head and a pre-trained CNN, ResNet50 architecture is applied in the second phase, model development, fine-tuned for fruit categorisation. Six groups of images are classified: fresh peaches, fresh pomegranates, fresh strawberries, rotten peaches, rotten pomegranates, and rotten strawberries. The third phase training and validation focuses on training the model; the remaining 15% is utilized for validation to maximize the

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hyperparameter values and stop overfits; with 70% of the dataset Using 15% of the dataset left aside for independent testing, the last phase, assessment assesses the model. One measures the model using among other evaluation metrics accuracy, precision, recall, and F1-score. This method guarantees a thorough and strict evolution of a useful fruit categorization system.

Fine-Tuned ResNet50:

Designed to address challenging photo classification problems, ResNet50—often known as Residual Network with 50 layers—is a deep convolutional neural network. Its main finding is residual learning, in which the model can escape some layers using shortcut connections, therefore addressing the "vanishing gradient" problem sometimes impeding the training of deep networks. Beginning with an input layer of a 7x7 convolutional layer with 64 filters and a stride of 2 employing a 3x3 max pooling layer, the design lowers spatial dimensions. ResNet50 consists on four basic phases, each with multiple residual blocks. Following the first stage, which consists of three residual blocks including 128, 128, and 512 filters in each block, the second stage, Conv3_x, consists of three convolutional layers (64, 64, and 256). In the third stage, or Conv4_x, six residual blocks apiece with 256, 256, and 1024 filters, so extracting even higher-level patterns from the images. Conv5_x is the penultimate phase of three residual blocks having 512, 512, and 2048 filters. This improved the complex features for suitable classification. The last classification follows the convolutional layers by using a global average pooling layer downing the feature mappings to a single vector of 2048 values. Excellent feature extraction capacity; this architecture is perfect for massive image datasets and has shown amazing performance in usage including fruit classification.

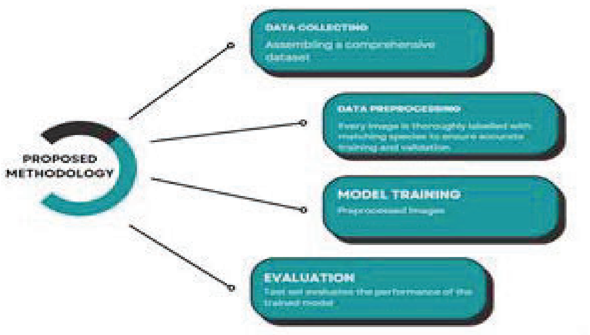

Fig. 2. Proposed Methodology.

V. RESULTS

With an overall accuracy of 95%, ResNet50 was good in identifying fresh and rotting fruit. Accuracy, recall, and f1-scores are strong for most categories; fresh_pomegranates_done and rotten_strawberries_done show excellent performance. The model demonstrates good generalizing with modest overfitting even if recall for rotten_peaches_done substantially drops. High values on the diagonal ensure accurate predictions using which the confusion matrix verifies effective classification. These results support the consistency of the model and its possible application in fruit classification.

A. *Classification Report Analysis*

The full evaluation of six classes fresh_peaches_done, fresh_pomegranates_done, fresh_strawberries_done, rotten_peaches_done, rotten_pomegranates_done, and rotten_strawberries_done reveals in the ResNet50 classification report. With a usual accuracy of 95%, the model performs rather well in separating fresh from rotting fruits. Fresh_pomegranates_done earns the best f1-score of 0.99, however, each class's accuracy, recall, and f1-score are typically really good. Likewise showing great performance with f1-scores of 0.95 or above are fresh_peaches_done, fresh_strawberries_done, rotten_pomegranates_done. The model's overall f1-score (0.91) in this class declines even if its recall (0.87) for rotten_peaches_done is rather lower. Reflecting the general balanced performance among all classes. While some categories like rotten_peaches_done show space for development—these results imply that the ResNet50 model finds fresh from rotting fruit photos rather successfully. For the dataset, the model produces rather high general accuracy. As shown in Fig.3.

	precision	recall	f1-score	support
fresh_peaches_done	0.91	1.00	0.95	41
fresh_pomegranates_done	0.98	1.00	0.99	47
fresh_strawberries_done	0.91	0.97	0.94	31
rotten_peaches_done	0.96	0.87	0.91	55
rotten_pomegranates_done	0.95	0.95	0.95	38
rotten_strawberries_done	0.97	0.92	0.94	37
accuracy			0.95	249
macro avg	0.95	0.95	0.95	249
weighted avg	0.95	0.95	0.95	249

Fig. 3. Classification Report Analysis

B. *Training and Validation loss Analysis*

The performance of the ResNet50 model in the classification challenge mostly relies on the training and validation losses. Since the model falls consistently across the training process, its loss implies that it is learning to differentiate the fresh from rotting fruit categories. A minimal variation in the training and validation loss indicates that the model does not overfit the training set but rather generallyizes effectively to raw data. Under this situation, the stabilisation of the validation loss near to the training loss as training advances offers a favourable sign of the capacity of the model to sufficiently forecast the classification of fruits from the validation set. Consistent falling loss over training in conjunction with the stable validation loss reveals the durability of the ResNet50 model. Moreover underlining its accuracy and reliability in projecting fresh and rotten fruits over numerous classes, the model may lower errors while maintaining a solid fit to both training and validation datasets. This clarifies the high classification results including the 95% general accuracy. As shown in Fig.4.

C. *Training and Validation Accuracy Analysis*

Training and validation accuracy help to explain how well the ResNet50 model detects fresh from decaying apples. Reflecting its higher ability to categorize the fruit images it was trained on, the model exhibits a continuous improvement in training accuracy during the training duration. This increasing trend suggests that the model is learning and adapting to fit the properties of the several fruit groups pretty

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successfully. Not less significant is the model's rather similar validation accuracy to its training accuracy. The great validation accuracy ensures that the performance of the model is strong and constant since the model appropriately detects the fruit images in the validation set with low overfitting or bias. The general accuracy of 95% on the test set supports the model's exceptional performance even further. This remarkable accuracy indicates that among the six varieties of fresh and decaying apples, the ResNet50 model can distinguish them rather successfully. The consistency between training and validation accuracy highlights the capacity of the model to generalize well, hence it is a trustworthy instrument for beneficial purposes in fruit categorization. As shown in Fig.5.

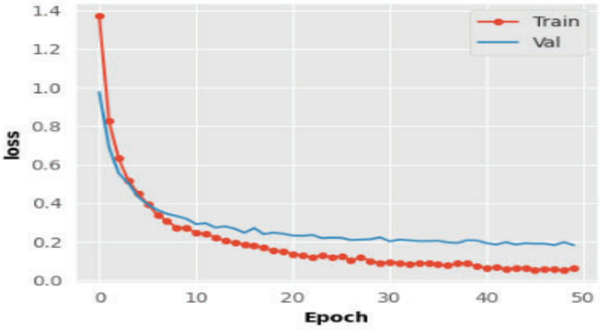

Fig. 4. Training and Validation loss Analysis

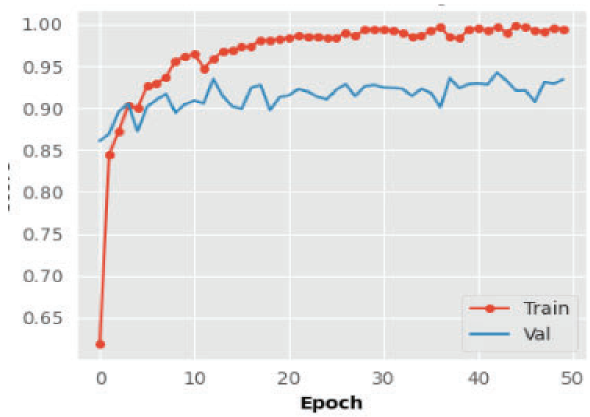

Fig. 5. Training and Validation Accuracy Analysis

D. Confusion Matrix

The confusion matrix offers the entire picture of ResNet50 model test set classification performance. Freshpeaches done, fresh_pomegranates done, fresh_strawberries done, rotten_peaches done, rotten_pomegranates done, and rotten_strawberries done Diagonal elements also aid in correct classifications and illustrate the accuracy of the model in many fruit varieties. While high values along the diagonal imply effective categorization, off-diagonal levels expose misclassifications. The heatmap correctly displays the performance of the model with comments and colour shading, therefore stressing areas for improvement and projecting fruit types. Using identification of particular difficulties in categorisation and validation of the general correctness of the model, this matrix leads future advances to raise the prediction capacity of the model. As shown in Fig.6.

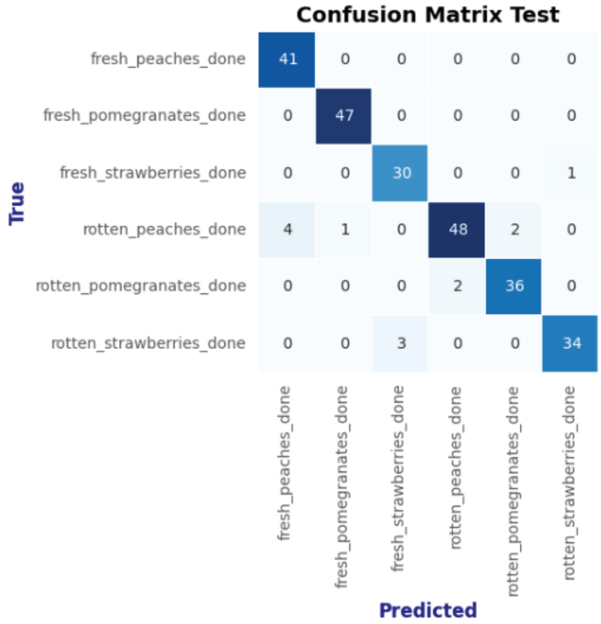

Fig. 6. Confusion Matrix

VI. Conclusion

This work presents the fresh from rotten apple difference of the ResNet50 model with an overall accuracy of 95%. Particularly effective at separating fresh pomegranates from bad strawberries, over numerous evaluation criteria—including accuracy, recall, and f1-score the model has displayed good performance. Based on the training and validation assessments, the ResNet50 model demonstrates high generalisation with little overfitting and performs regularly over the datasets. The confusion matrix highlights, even more, the model's fruit-identifying accuracy as well as some areas requiring development. The results show how effectively automated fruit classification systems may reduce food waste in areas of food and industry and enhance quality control. Using state-of-the-art computer vision algorithms like ResNet 50, this work presents a scalable system to monitor fruit freshness, improve supply chain efficiency, and assist sustainable food management. Future research should concentrate on improving the model to enable even more accuracy and spreading its applicability over a wider range of fruits and surroundings of the environment.

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